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Accounting for positional uncertainty in historical shoreline change analysis without ground reference information

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ABSTRACT

Systematic shifts in shoreline position are important indicators of environmental change. Shoreline position interpreted from historical aerial imagery is frequently used to assess shoreline change. Although most published studies do not formally consider the effects of source error and interpretation error when analysing shoreline change, the effects of these errors may be significant. This article introduces and evaluates a new uncertainty-aware approach to assessing shoreline change in the presence of positional uncertainty and without ground-reference data, which is typical for historical coastline analysis. The overlapping double buffer (ODB) approach extends the epsilon band model to account for the effects of source and interpretation errors on shoreline position, and assesses the degree of overlap in order to distinguish between significant change and noise. This approach improves upon standard shoreline change analytical techniques that use regularly spaced shore-normal transects to measure the direction and magnitude of shoreline change without accounting for error. Shoreline position interpreted from historical aerial images for four sites along the Michigan (USA) coast between 1938 and 2010 were used to demonstrate the feasibility of the ODB approach. An epsilon band was constructed around each shoreline with radius equal to the combined source and interpretation error for each image. These bands were merged and intersected to test whether the observed change was real or an artefact caused by uncertainty in the data sources. The most significant advantage of the ODB approach is its ability to account for uncertainty in shoreline position where ground-reference information does not exist, as is often the case with historical aerial imagery. Results indicate that the overlapping epsilon band method is viable for analysing change in linear features in the absence of a higher-accuracy reference dataset, and that source error contributed more to the positional uncertainty than did interpretation error.

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1. Introduction

Coastal erosion is an important problem facing many coastal environments and communities. One common approach to assessing coastal erosion and deposition is by measuring the horizontal movement of the shoreline position. The accurate assessment

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of shoreline change is recognized by the U.S. Geological Survey (USGS) as an issue of great importance that is vital to better understanding large-scale and fine-scale patterns in coastal environments (Hapke et al. 2006; Hapke, Malone, and Kratzmann 2009; Hapke et al. 2010). Elko et al. (2016) recently highlighted accurately identifying long-term shoreline change rates as an important research need for understanding dune morphology and resiliency along developed coasts. Long-term patterns of shoreline change are commonly assessed using historical aerial imagery, often times without ground-reference information about the true shoreline position at the time the image was acquired. Current approaches to assessing patterns of change require ground-reference information or assumptions about the magnitude and direction of error (i.e. Skidmore and Turner 1992; Goodchild and Hunter 1997; Tveite 1999; Heuvelink, Brown, and VanLoon 2007; Mozas and Ariza 2010). In addition to fieldwork, geospatial methods are increasingly used to assess past and present changes using a combination of historical charts, historical aerial imagery, and lidar.

An important aspect of shoreline change assessment and an emerging theme in geographic information science (GIS) is the incorporation of uncertainty assessment in spatial analysis. The positional accuracy of a digital shoreline can be determined by the epsilon band method (Skidmore and Turner 1992), the simple buffer method (Goodchild and Hunter 1997), the double buffer method (Tveite 1999), a combination of these methods (Mozas and Ariza 2010), or estimating multivariate joint probability distribution functions (e.g. Heuvelink, Brown, and VanLoon 2007). However, these approaches require 'data of higher geometric accuracy' (Tveite 1999, 29) as reference data in order to assess the accuracy of historical remotely sensed data, as is often not the case with shoreline change assessments based on historical aerial imagery. While this error may be significant, current methods cannot be used to determine how these errors affect the results of the shoreline change assessment. As a consequence, comprehensive error quantification and propagation remain absent or limited in many shoreline change studies (e.g. Ruggiero, Kaminsky, and Gelfenbaum 2003; Hapke et al. 2006; Hapke, Malone, and Kratzmann 2009; Hapke et al. 2010).

Accurately measuring shoreline change first requires that we understand what affects the apparent shoreline position at any given point in time. Shoreline position will vary naturally by fluctuations in the water-plane elevation, changing wave conditions, along-shore and onshore sediment transport, beach nourishment engineering projects, and other natural and anthropogenic processes. The apparent shoreline position may suffer from positional uncertainty due to data source error, interpretation error, and natural variability (Ruggiero, Kaminsky, and Gelfenbaum 2003; Heuvelink, Brown, and VanLoon 2007; Sutherland 2012). Source error is caused by inaccuracy in the data source, such as the positional uncertainty of the modern image that is used to georeference the historical aerial imagery is being referenced (Ruggiero, Kaminsky, and Gelfenbaum 2003). In addition to source error, interpretation error (i.e. error due to difficulty identifying and digitizing shoreline position from historical aerial imagery) also affects the apparent shoreline position (Sutherland 2012). The method used to calculate source of error will depend greatly on the data source and technique used to extract the feature. Contemporary shoreline change studies have not quantitatively estimated and accounted for these different sources of error.

Error due to natural variability is caused by fluctuations in environmental conditions that cause the shoreline to appear in different locations. Shoreline position is affected by the relative tidal range and seasonal water plane fluctuations (Brinkmann 2000; Lenters 2001; Quinn 2002; Lenters 2004; Sellinger et al. 2008) and will be most variable on macrotidal coasts and least variable on microtidal coasts. The Great Lakes coastlines are ideal locations to examine the effects of source and interpretation error because the lakes have a tidal range of only ~5 cm and are essentially non-tidal (NOAA National Ocean Service 2011). Effects of seasonal water level fluctuations on the apparent shoreline location can be mitigated by using imagery from the same time of year. The effects of source error, interpretation error, and natural variability are often overlooked, but can result in significant differences in apparent shoreline position. This article focuses on measuring and modelling the effects of source and interpretation error on characterizing shoreline change.

The purpose of this article is to evaluate the utility of coupling spatially continuous shoreline representation with uncertainty propagation modelling to better measure the significance of calculated shoreline change where high accuracy reference data are not available. In particular, we refine the use of epsilon bands for detecting change in shoreline position through time along a non- to micro-tidal coast. In any particular application, the special physical properties of the shoreline position (i.e. shoreface slope, beach slope, wave run-up) and the drivers for the change (i.e. normal daily wave action, extreme storm event) may be important in order to understand the system as a whole; however, we focus here solely on generic questions associated with shoreline comparison. We introduce a new approach for analysing shoreline change in the presence of positional uncertainty and without high accuracy reference information. This overlapping double buffer (ODB) approach can test whether the observed positional difference between features is conflated with error. In addition, this article provides insight into the relative influence of different types of error (e.g. georeferencing error, interpretation error) on shoreline change analyses. The utility of the new approach is demonstrated using shorelines interpreted from aerial imagery at four sites along the west coast of Michigan's Lower Peninsula, USA (Figure 1).

1.1. Current approaches to assessing changes in line objects

A variety of methods have been proposed to assess the positional accuracy of geographic features (Mozas and Ariza 2010). In order to assess the accuracy of a particular feature, it is important that the quality of a reference dataset is known first (Skidmore and Turner 1992; Tveite and Langaas 1995, Tveite 1999). Assuming that this condition is met, the accuracy of the line being tested can be determined by the Hausdorff distance method (Abbas et al. 1995), the epsilon band method (Skidmore and Turner 1992), the simple buffer method (Goodchild and Hunter 1997), the double buffer method (Tveite 1999), or a combination of these methods (Mozas and Ariza 2010). The two commonly applied methods are those of Skidmore and Turner (1992) and Tveite and Langaas (1995; Tveite 1999), which employ epsilon bands to characterize positional uncertainty and assess the accuracy of a particular feature.

Skidmore and Turner (1992) outlined a method for assessing the accuracy of linear features that uses a combination of epsilon bands and intersecting line segments.

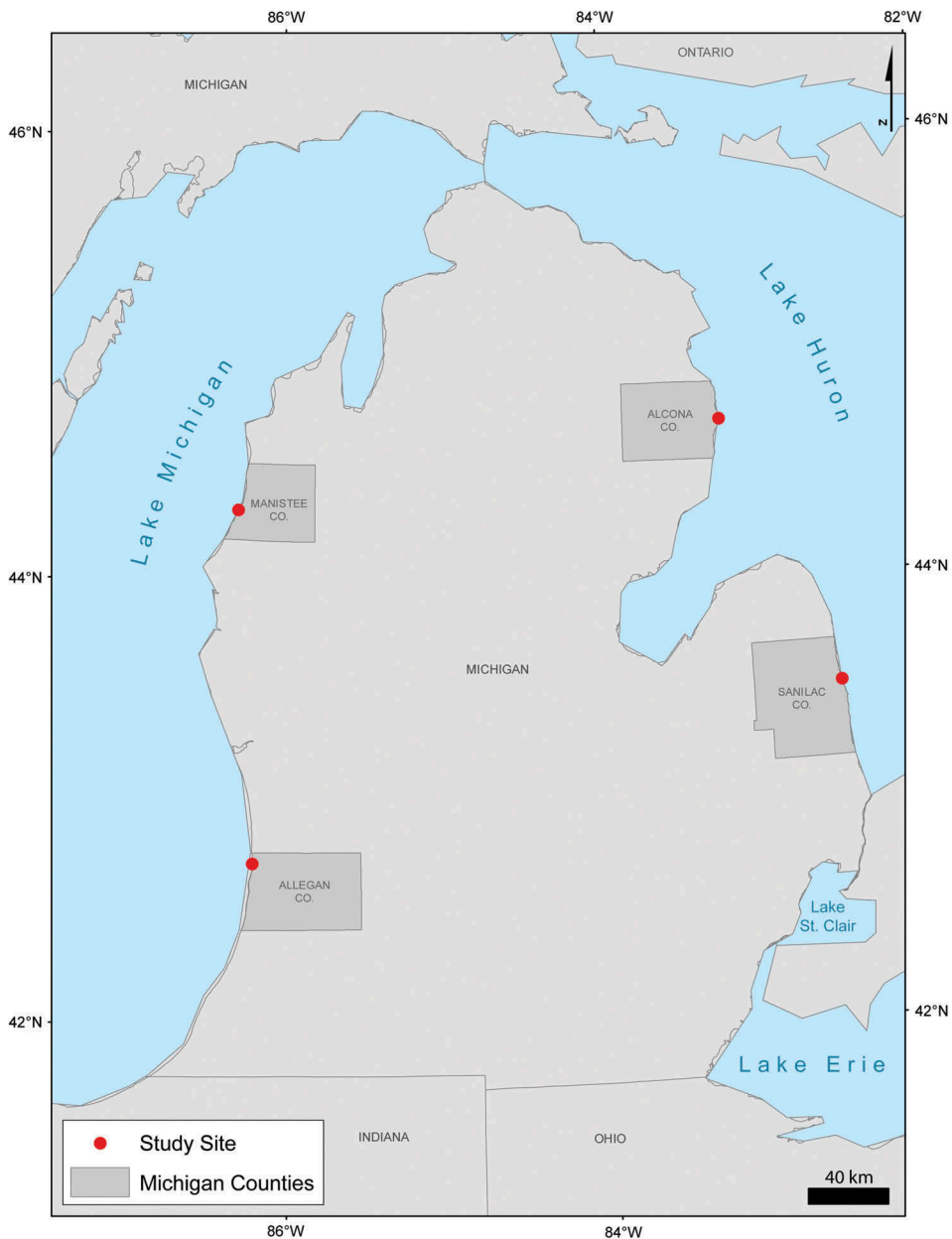


Figure 1. Study sites used to demonstrate the feasibility of the ODB approach were located in four quadrants of Michigan's Lower Peninsula.

Another method for assessing accuracy of lines was outlined by Tveite and Langaas (1995; Tveite 1999), where buffers around lines of known (presumably higher) accuracy and lines of unknown accuracy are used to calculate the accuracy of the line with the unknown accuracy. Both of these methods require high-accuracy reference information about the position of the feature, which is often not present for historical charts, maps, and aerial imagery.

2. Methods

2.1. ODB model

In the ODB approach, a measure of positional uncertainty is calculated for the shorelines derived in each image. Then, epsilon bands are used to assess the probability that the observed change is real and not due solely to source error and interpretation error. Finally, since buffers are not able to provide a spatially variable measure of distance between shorelines (i.e. distance between shorelines is not likely to be spatially homogeneous), regularly spaced shore-normal transects are employed to compute the direction and magnitude of apparent shoreline change (Figures 2 and 3). Each of these steps is detailed in this section.

2.2. Estimating positional uncertainty

We characterize positional uncertainty in shoreline location as having three independent sources: positional error in the historic imagery, georeferencing error, and interpretation error (Equation (1); Sutherland 2012). Root mean squared error (RMSE) is used to summarize the error contributed by each of these sources. RMSE is a standard metric for characterizing the average deviation of observed positions from actual positions. $RMSE_B$ represents the RMSE of the base image to which the historical image is referenced (extracted from the imagery metadata), and $RMSE_G$ represents the RMSE introduced by the georeferencing process. Shoreline position is subject to interpretation uncertainty, as photo interpreters will not agree on the exact shoreline position in a given image (Edwards 1999; Zhang and Goodchild 2002). These differences can be quantified by identifying shoreline position on a series of shore-normal transects. The mean position of the shoreline on each transect can be calculated, as well as the squared distance d^2 of each of interpreted position from this mean. Assuming interpretations are unbiased, interpretation error is estimated from the n sample shoreline locations across all transects as

$$RMSE_I = \sqrt{\sum d^2/n}. \quad (1)$$

The overall positional RMSE is the square root of the sum of squared RMSE values from independent sources (Equation (2); Sutherland 2012). In accordance with the US National Standards for Spatial Data Accuracy (NSSDA) specifications, a more appropriate horizontal accuracy standard is $RMSE_{95}$, which is calculated by multiplying the cumulative RMSE by 1.7308, where errors are assumed to be equal in all directions (i.e. radial case; Equation (3); FGDC 1998). Assuming that errors are unbiased, independent, and bivariate normally distributed, $RMSE_{95}$ indicates the distance below which 95% of the positional errors in the image are expected to fall (Greenwalt and Schultz 1968; Congalton and Green 2009; $RMSE_{95}$; Table 1):

$$RMSE_O = \sqrt{RMSE_B^2 + RMSE_G^2 + RMSE_I^2}, \quad (2)$$

$$RMSE_{95} = 1.7308 \times RMSE_O, \quad (3)$$

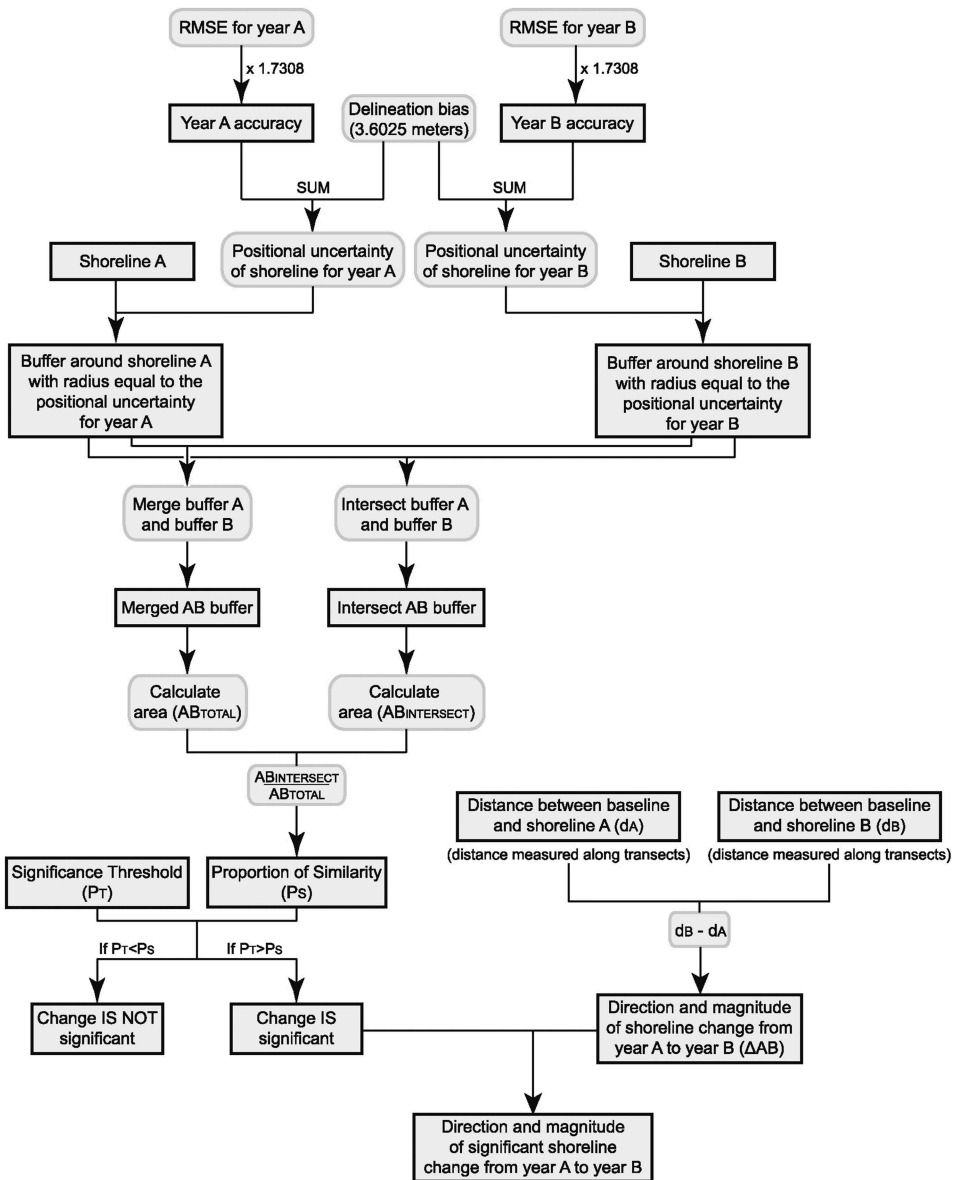


Figure 2. Geoprocessing model used to identify the significance, direction, and magnitude of shoreline change from one year to another. Boxes with dark borders and square corners indicate GIS objects or key values, and boxes with light borders and rounded corners represent processes.

where $RMSE_O$ is the overall RMSE of the shoreline, $RMSE_B$ represents the RMSE of the base image to which the historical image is referenced, $RMSE_G$ represents the RMSE introduced by the georeferencing process, and $RMSE_I$ represents the interpretation error. $RMSE_{95}$ and $RMSE_{BG95}$ are the 95% error of the shoreline positions with and without accounting for interpretation error, respectively.

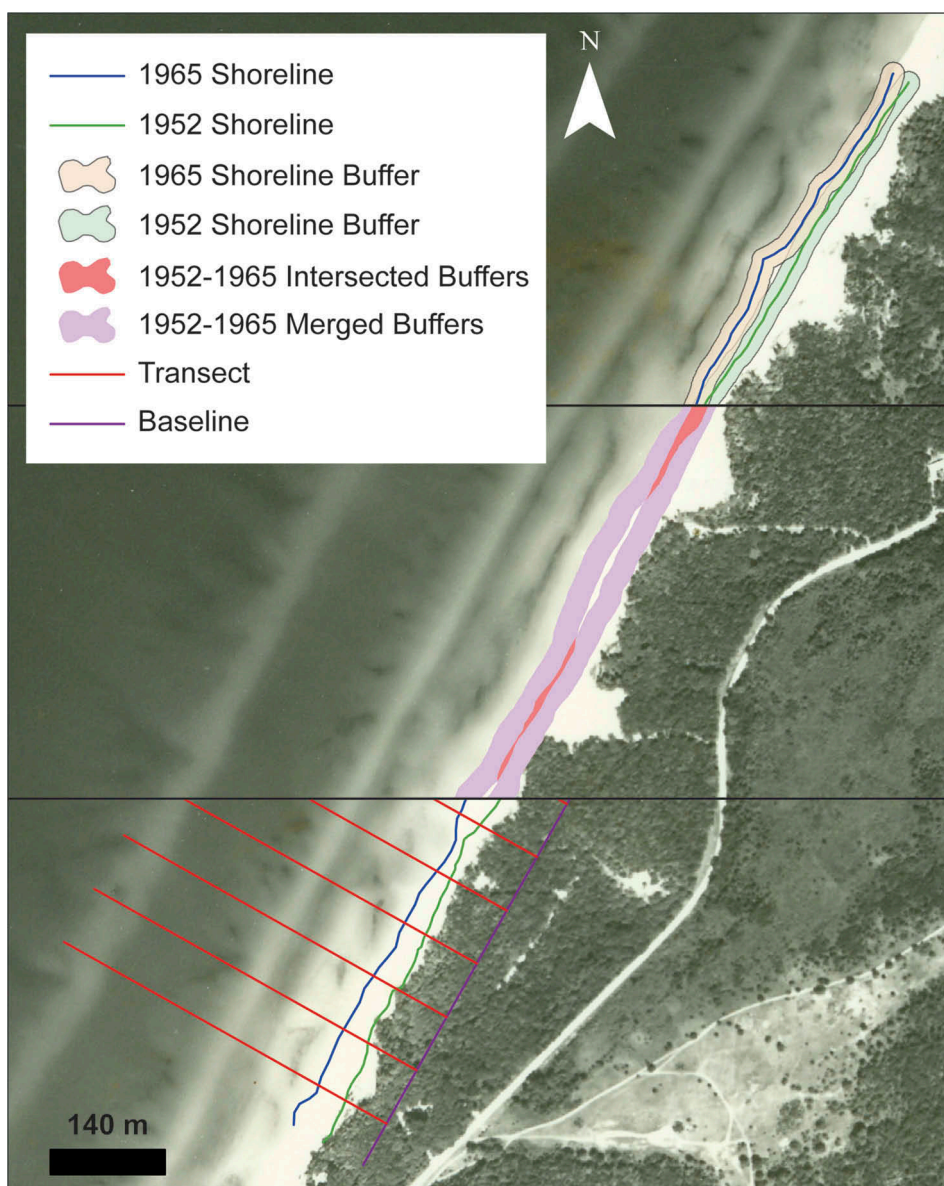


Figure 3. Use of dual epsilon bands to analyse shoreline change. (Top) The first step of the dual epsilon band analysis is to generate a buffer around each shoreline with a radius equal to its uncertainty. (Middle) The second step of the dual epsilon band analysis is to merge the two buffers and calculate its area (purple region). The area of intersection of the two buffers (red regions) is also calculated. (Bottom) Transects were used to determine the direction and magnitude of shoreline change, following standard approaches.

2.3. ODB

A buffer is generated around shoreline at times t_0 and t_1 with radius equal to shoreline accuracy (A) for t_0 and t_1 , respectively (e.g. blue and light red areas in Figure 3). These buffers are merged ($A_1 \cup A_2$; e.g. purple area in Figure 3) and

Table 1. Calculated accuracy of historical aerial imagery used to extract shoreline position. RMSE_B corresponds to the accuracy of the image to which the historical imagery was georeferenced.

	Year	No. of GCPs	RMSE _B (m)	RMSE _G (m)	RMSE _I (m)	RMSE _{BG} (RMSE _{BG95}) (m)	RMSE _O (RMSE ₉₅) (m)
NW site	1938	11	6.00	2.46	2.53	6.48 (11.22)	6.96 (12.05)
	1952	10	6.00	1.74	2.53	6.25 (10.81)	6.74 (11.67)
	1965	11	6.00	1.94	2.53	6.31 (10.91)	6.79 (11.76)
	1973	10	6.00	1.89	2.53	6.29 (10.89)	6.78 (11.74)
	1992	–	10.15	–	2.53	10.15 (17.57)	10.46 (18.11)
	1998	–	10.15	–	2.53	10.15 (17.57)	10.46 (18.11)
	2005	–	10.15	–	2.53	10.15 (17.57)	10.46 (18.11)
	2009	–	6.00	–	2.53	6 (10.38)	6.51 (11.27)
SW site	2010	–	6.00	–	2.53	6 (10.38)	6.51 (11.27)
	1938	12	6.00	1.67	2.53	6.23 (10.78)	6.72 (11.64)
	1950	10	6.00	1.60	2.53	6.21 (10.75)	6.71 (11.61)
	1960	10	6.00	2.09	2.53	6.35 (11)	6.84 (11.84)
	1967	11	6.00	1.86	2.53	6.28 (10.87)	6.77 (11.72)
	1974	10	6.00	2.78	2.53	6.61 (11.45)	7.08 (12.25)
	1998	–	10.15	–	2.53	10.15 (17.57)	10.46 (18.11)
	2005	–	10.15	–	2.53	10.15 (17.57)	10.46 (18.11)
NE site	2009	–	6.00	–	2.53	6 (10.38)	6.51 (11.27)
	2010	–	6.00	–	2.53	6 (10.38)	6.51 (11.27)
	1938	10	6.00	0.92	2.53	6.07 (10.51)	6.58 (11.38)
	1952	10	6.00	1.45	2.53	6.17 (10.68)	6.67 (11.55)
	1963	10	6.00	2.74	2.53	6.6 (11.42)	7.06 (12.23)
	1979	12	6.00	2.51	2.53	6.5 (11.26)	6.98 (12.08)
	1992	–	10.15	–	2.53	10.15 (17.57)	10.46 (18.11)
	1998	–	10.15	–	2.53	10.15 (17.57)	10.46 (18.11)
SE site	2005	–	10.15	–	2.53	10.15 (17.57)	10.46 (18.11)
	2009	–	6.00	–	2.53	6 (10.38)	6.51 (11.27)
	2010	–	6.00	–	2.53	6 (10.38)	6.51 (11.27)
	1941	10	6.00	0.68	2.53	6.04 (10.45)	6.55 (11.33)
	1955	12	6.00	1.39	2.53	6.16 (10.66)	6.66 (11.52)
	1963	10	6.00	0.6	2.53	6.03 (10.44)	6.54 (11.32)
	1982	12	6.00	2.6	2.53	6.54 (11.32)	7.01 (12.14)
	1998	–	10.15	–	2.53	10.15 (17.57)	10.46 (18.11)
	2005	–	10.15	–	2.53	10.15 (17.57)	10.46 (18.11)
	2009	–	6.00	–	2.53	6 (10.38)	6.51 (11.27)
	2010	–	6.00	–	2.53	6 (10.38)	6.51 (11.27)

RMSE_G refers to the computed accuracy of the georeferencing process (based on a network of ground control points), and RMSE_I refers to the interpretation accuracy measured using a series of shore–normal transects. The RMSE_B, RMSE_G, and RMSE_I are combined to compute the RMSE_O, which is used to compute the spatial accuracy (at 95% confidence level) of the historical georeferenced image (RMSE₉₅). Years without an RMSE_G value were available digitally, in which case, RMSE_O is computed using RMSE_B and RMSE_I. RMSE_{B95} represents positional uncertainty excluding interpretation error.

intersected ($A_1 \cap A_2$; e.g. bright red area in Figure 3). The proportion of similarity (P_S) is computed by dividing the intersected area by the merged area (Equation 4), and the resulting value, ranging from 0 to 1, represents the proportion of overlap from t_0 to t_1 . High values suggest that the measured change is confounded by uncertainty; whereas, low P_S values suggest that the measured change is real. The calculated P_S value is compared to a user-defined significance threshold (T) to test whether the observed change is significant:

$$P_S = \frac{A_1 \cap A_2}{A_1 \cup A_2}, \quad (4)$$

$$P_S < T, \quad (\text{Case1})$$

$$P_S \geq T. \quad (\text{Case2})$$

If case 1 is determined, the features are judged to be in significantly different locations, and therefore that the shoreline has changed. Conversely, if case 2 is determined, then measured differences in feature position are not sizable enough to be distinguishable from positional uncertainty. Choice of T is clearly important: low T values, such as 0.05 and 0.10, are more conservative, with lower likelihood of resulting in errors of commission (incorrectly identifying significant shoreline change), but higher likelihood of errors of omission (failing to identify actual shoreline change).

2.4. Transect sampling

It is unlikely that the distance between shoreline position at two different times is spatially homogenous; therefore, it would be inappropriate to compute distance between shoreline positions using a buffer approach. Instead, regularly transect–intersect sampling is applied to compute the magnitude and direction of shoreline change along shore–normal transects (Figures 3 and 4; e.g. Anders and Byrnes 1991; Houser, Hapke, and Hamilton 2008; Barnhardt 2009; Hapke, Malone, and Kratzmann 2009). This approach is analogous to the widely applied Digital Shoreline Analysis System (DSAS) GIS utility, which was developed by the USGS to assess shoreline change. It is important to note that transect–intersect sampling is only used to determine direction and magnitude of change, not significance of the observed change. The distance between a pre-defined baseline and each shoreline–transect intersection was used to compute direction and magnitude of change from t_0 to t_1 (Figures 2 and 3). The four sites utilized in this article all have relatively linear coastlines and the analysis utilized a linear baseline; however, it is possible that this methodology could be

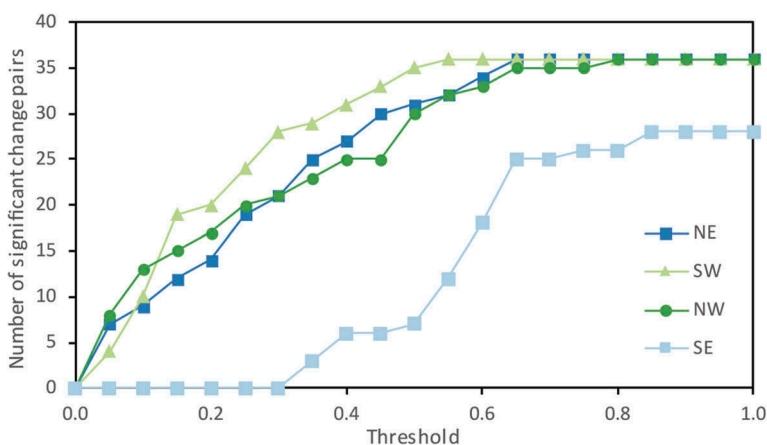


Figure 4. Relationship between the significance threshold and the number of pairs of years with significantly different shoreline positions. The importance of establishing an appropriate threshold value is highlighted by the differences between the NE, SW, and NW sites and the SE site. Lower thresholds are more conservative measures of change, but are more likely to result in errors of omission and may underestimate the significance of change from one time to another. Higher threshold values are more liberal measures of change and more likely to include errors of commission than omission.

employed in locations where the shoreline has a significant curve. In cases where the coast is significantly curvilinear, a curved baseline would be more appropriate. Negative values indicate that coastal erosion has occurred; whereas positive values indicate that coastal accretion has occurred.

2.5. Additional methodological considerations

The ODB approach assumes that (1) shorelines are independently interpreted, (2) positional uncertainty can be estimated for all shorelines, (3) the calculated uncertainty represents a conservative measure of accuracy, which is constant along the entire length of the shoreline.

The assumption that shoreline observations are independently delineated is frequently made in coastal change studies (Thieler and Danforth 1994; Pendleton, Thieler, and Williams 2005). To ensure that a shoreline segment is independent, no other shorelines should be visible during the interpretation process. This minimizes the likelihood that the position of the shoreline from onetime frame will influence the interpreted shoreline position from another time frame.

It is important that positional uncertainty of every shoreline can be calculated in order to determine whether the observed change is real or confounded with uncertainty. The historical shorelines used in this research were determined by visual interpretation of georeferenced aerial photographs. During the georeferencing process, RMSE was calculated for each individual photograph, which was combined with the RMSE of the reference image (USDA 2011).

3. Data sources

The shorelines analysed for this article's case study were interpreted from historical aerial imagery, which was acquired at several points in time between 1938 and 2010 (Table 1) along the coast of Lower Michigan. This is a standard and relevant application, as interpretation of shoreline position from air photos is well documented (e.g. Moffitt 1969; Stafford and Langfelder 1971; Seibel 1972; Hands 1979, 1980; Anders and Byrnes 1991; Pendleton, Thieler, and Williams 2005). Historical airphotographs used from which shoreline position was interpreted from were selected based on availability, image type (i.e. greyscale, true colour), photo scale, and image contrast.

This research used images that were taken at all four sites during the same month (July), in order to minimize the likelihood of short-term impact by large storms with high winds and waves, which are rarest during July. Imagery was either greyscale or true colour and had a photo scale of 1:24,000 in order to ensure that the interpreted shoreline position was based on a consistent set of image characteristics at a consistent level of detail. Finally, since coastal imagery is often overexposed due to the highly reflective bare sand, historical images used in this case study could not be under- or over-exposed.

3.1. Source error

When possible, the overall image accuracy ($RMSE_i$) was extracted from the digital imagery metadata. Pre-1992 imagery was only available in hard copy form and had to be scanned at high resolution (at least 600 dpi) and georeferenced in a GIS. The resulting

digital images had a 1 m resolution, and were georeferenced to the 2010 National Agriculture Imagery Program (NAIP) imagery using ArcGIS 10®. A minimum of 10 geographically distributed ground control points (GCPs), consisting of low-elevation stable features with minimal radial distortion (e.g. persistent road intersections), were used to reference the scanned historical images to the 2010 imagery. The maximum RMSE among all the georeferenced images was less than 3 m (Table 1); and since the historical aerial imagery was georeferenced to the 2010 NAIP imagery, the 6 m accuracy of the 2010 imagery (USDA 2011) was included in the cumulative RMSE.

Imagery was georeferenced instead of orthorectified because the process of orthorectification requires a digital elevation model (DEM) of the same time period, which does not exist for many historical images. Since there is no time-matching DEM, orthorectification of many historical images is not possible. Although a modern DEM could be used for orthorectification, the orthorectified images would not be accurate representations because the landscape has changed between when the imagery was acquired and when the DEM data was collected.

3.2. Interpretation error

Shorelines were interpreted from imagery based on tonal contrast at the land–water interface. The high moisture content of the wetted beachface sands decreases its spectral reflectivity (Crowell, Leatherman, and Buckley 1991; Boak and Turner 2005; Li, Di, and Ma 2005) and causes the wetted beachface to appear darker than the dry beach. This change occurs over a narrow zone known as the high water line (HWL), which can be visually interpreted, and is the best indicator of the shoreline position for historical shoreline change studies, particularly in non-tidal areas (Crowell, Leatherman, and Buckley 1991; Boak and Turner 2005).

In order that the uncertainty analyses include the possible interpreter errors in the base shoreline delineations, two other remote-sensing professionals independently delineated the shoreline for three historical air photos, resulting in six independently interpreted shorelines (three historical images for each of the two remote-sensing professionals). Using 20 shore–normal transects, the independently interpreted shorelines were compared and the mean interpretation error was estimated to be 0.66 ± 1.47 m. Interpreted shoreline positions deviating from the mean interpreted shoreline position along a transect by 20 m or more are considered ‘blunders’ and excluded from further analyses. Throughout the rest of this discussion, a single pair of times is referred to as a change pair, and a pair of times with significantly different shoreline positions is referred to as a significant change pair.

4. Results

4.1. Transect–intersect and ODB

Shoreline position was highly dynamic at three of the sites (NW, SW, and NE sites), but was relatively stable at the SE site. The NW site had the largest number of significant change pairs at the three most conservative significance thresholds (Table 2). Conversely, the SE site had the fewest pairs of years with significantly different shoreline position among the four test sites (Figure 4).

The number of significant change pairs varied by site (Figure 4) and were positively correlated with the threshold value at each site. This relationship was strongest at the

Table 2. Summary table of results with key parameters for all four sites.

	NW site	SW site	NE site	SE site
	Sand	Sand	Sand	Sand, gravel, cobbles
Dominant particle size				
Range in shoreline position (m)	53.38	40.44	51.23	15.15
Endpoint change (m)	−30	−10.9	15.17	−5.27
Average annual change (endpoint rate) (m year ^{−1})	−0.41	−0.15	0.21	−0.07
Lake level – shoreline position relationship (coefficient of variation; R^2)	0.44	0.24	0.81	0.83
Minimum threshold with ≥ 1 significant change pair	0.05	0.05	0.05	0.35
Notes	None	Tall bluffs (>5 m)	None	Groins

Sieving and laser-diffraction were used to determine the dominant particle size. Range in shoreline position, endpoint change, average annual change, relationship between lake level and shoreline position, minimum spatial autocorrelation, and minimum threshold with at least 1 significant change pair were determined using a combination of ArcGIS® and R. Average annual change was computed using an endpoint change rate, where the total magnitude of change is divided by the number of elapsed years. Notes highlights some of the important characteristics of each site based on field notes collected during summer 2011.

NW and NE sites, and was weakest at the SE site. The weaker correlation evident at the SE site is likely due to the fact that the first significant change pair did not appear until a much higher threshold value (0.35) compared to the other three sites (Table 2). In addition to the high threshold, there was one less sample year at the SE site, which reduced the number of possible temporal pairwise combinations.

4.2. Relative importance of errors

The average error contributed by image processing and georeferencing was approximately $12.69 \pm 3.14 \text{ m}^2$, and the error contributed by shoreline interpretation was $13.46 \pm 3.00 \text{ m}^2$. There was a statistically significant ($p < 0.001$) difference between the amount of error contributed by imagery already in digital format (i.e. post-1992 imagery; mean $\text{RMSE}_{95} = 15.07 \pm 3.49 \text{ m}^2$) and the amount of error contributed by scanned and georeferenced analogue imagery (mean $\text{RMSE}_{95} = 11.75 \pm 0.30 \text{ m}^2$).

In contrast to the high amount of error solely due to image processing (mean $\text{RMSE}_{895} = 12.69 \pm 3.14 \text{ m}^2$), the amount of error contributed by shoreline interpretation and delineation was relatively small (mean $\text{RMSE}_{95} = 13.46 \pm 3.00 \text{ m}^2$). Although the interpretation error was calculated using imagery for three sample years at only one site, the imagery for these three sample years were the most difficult to interpret. As such, it is assumed that this statistic is conservative, and other interpreted shorelines may actually be more accurate than this calculated error.

5. Discussion

Shoreline position was least variable at the SE site and much of the observed change was found to be likely influenced by error. One possible explanation for the relative stability of this site compared to the three others is differences in the geologic setting coupled with effective fetch. This site is composed of larger sediments including gravels, cobbles, and boulders, which require substantially more wave energy to mobilize. Additionally, because the SE site is on the west coast of Lake Huron, the effective lake fetch of dominant westerly

winds is limited. This results in limited wave energy approaching the coast. In contrast, the other three sites are all composed of sandy sediments which can be moved with smaller waves. The SE site represents an important end member in beach morphology and is useful to examine how setting an appropriate threshold value is site- and application-dependent.

5.1. ODB approach

The ODB approach used in this study is a valuable tool to assess shoreline change from historical aerial imagery where ground-reference data is not available. This approach utilizes dual epsilon bands to determine whether the observed change is confounded with uncertainty, and uses shore-orthogonal transects to calculate the direction and magnitude of change (Figures 2 and 3). By combining these two methods, the ODB approach provides a measure of the direction and magnitude of change and the probability that the measured change is due solely to source and interpretation error, something previous shoreline change approaches do not account for.

Use of ODB for measuring historic shoreline change in the presence of positional uncertainty offers several advantages over conventional approaches. First, it propagates uncertainty from multiple sources through to study results: uncertainty associated with data collection and interpretation is used to determine the significance of the measured shoreline change. Previous studies acknowledge uncertainty in historical charts (Crowell, Leatherman, and Buckley 1991; Dolan, Fenster, and Holme 1991), ground surveys (Buckler and Winters 1983; LaMoe and Winters 1989), and aerial photographs (Gray and Wilkinson 1979; LaMoe and Winters 1989), but do not quantitatively estimate or propagate the consequent positional uncertainty through the change analysis process. By formally modelling contributions of different error sources, the ODB approach is able to inform an uncertainty-aware assessment of shoreline change.

Second, the ODB approach can be used across a wide variety of data sources. This study used shoreline positions interpreted from aerial imagery, but it is possible to apply the method to shorelines derived from other data sources such as ground surveys, GPS measurements, historical charts, or lidar. In many cases, the positional error of these sources can be quantified and then applied in the manner described in this article to characterize the positional uncertainty of the extracted shorelines.

Third, the ODB approach can be used to understand fine- and broad-scale patterns of change. Geographically broad spatial changes are identified by simply comparing S to T . Changes where T exceeds P_5 (case 1) are less significant than changes where P_5 exceeds T (case 2). Local, fine-scale changes are identified by the areas where the two buffers do not overlap at all (purple areas in Figure 3). Figure 4 highlights the importance of scale in establishing an appropriate threshold value (T) and how that threshold may vary between studies and across sites. In other words, the setting an appropriate threshold is site- and application-specific. Automating the selection of an appropriate threshold value or spatially variable threshold value is beyond the scope of this article and represents one way that future work can expand on the ODB approach presented in this article.

Finally, the most significant characteristic of the ODB approach is that it does not require independent reference shoreline information, which may often not be available for historical shoreline data. This study is typical in that no historical reference information about precise shoreline position is available for the times when the aerial images were acquired; it is

therefore not possible to assess the accuracy of our historical shorelines using methods introduced by Skidmore and Turner (1992) or Tveite and Langaas (1995; Tveite 1999). Instead, the epsilon band concept is adopted to construct a region of uncertainty around the represented shoreline. The epsilon band width is based on the positional uncertainty of the image and the uncertainty in shoreline interpretation. Further, we are not using epsilon bands simply to assess the spatial accuracy of the lines, but to assess the likelihood that the observed change in feature position is not conflated with error. Geosimulation approaches (e.g. Heuvelink, Brown, and VanLoon 2007) require either empirical information or assumptions about the spatial structure of the positional uncertainty, which requires a large amount of reference information. Without independent reference information about shoreline position, modelling the spatial structure of errors for geosimulation is not possible without making many untestable assumptions.

Because the significance of the change depends on the threshold magnitude, it is important to use an appropriately defined threshold (T). Higher T values are more liberal measures of change and more likely to include errors of commission: the incorrect identification of significant change, when little or no actual change has occurred. As T approaches 1.0, all change pairs may be identified as significant, effectively limiting the potential effects of uncertainty on shoreline position due to errors of omission. The appropriate threshold for a particular study will vary based on the research objectives and whether the researcher is more willing to accept errors of commission or errors of omission. Alternatively, assessing shoreline change across a range of T thresholds might be more illuminating. Objectively determining and automating the selection of an appropriate T threshold is beyond the focus of this study, which is simply meant to demonstrate the feasibility of the ODB approach to assess shoreline change when reference shoreline position information is not available.

5.2. Relative influence of sources of error

Shoreline position interpreted from historical aerial imagery suffers from two main types of error: source error and interpretation error. For the four sites reported here, source errors were greater than interpretation errors. Image processing contributed approximately 5.02 times more error than did interpretation error, while the amount of error was substantially different for imagery from different years. All imagery used in this study had either been previously georeferenced or was referenced to the 2010 NAIP imagery. The greatest source of error for scanned and georeferenced images was the pre-calculated accuracy of the 2010 imagery. Images that were scanned and referenced had a georeferencing RMSE of less than three metres, which is equal to one half of the 2010 imagery RMSE (6 m; USDA 2011). For this study, replacing the 2010 NAIP imagery with a higher-accuracy alternative would offer the greatest impact in reducing positional uncertainty, although no higher-accuracy alternative was available. These results are of course study-specific, but they suggest that analysts can use the methodology to isolate and reduce major sources of shoreline change uncertainty.

6. Conclusion

This study demonstrates the feasibility of the ODB approach to analyse shoreline change. The case study highlights this method using a time series of delineated shorelines at four sites in Lower Michigan (Figure 1). The advantages of this method are: (1) it quantitatively

estimates and propagates positional uncertainty, (2) it is applicable across a variety of data sources, and (3) it highlights areas that are more stable and areas that are more dynamic through time. Since the significance of the observed change is based on the proportion of similarity and a user-defined threshold, it is important that this threshold be appropriately defined according to the objectives and site(s) of the study.

In addition to demonstrating a new approach for assessing shoreline change, case study findings indicated that georeferencing errors contributed more error than did interpretation error. This is an important insight, since a large number of studies interpret shoreline position from aerial imagery and maps, but do not account for the positional accuracy of the source imagery/maps. Quantifying positional uncertainty and propagating its effect on change detection is demonstratively important for accurately assessing shoreline change and other applications. This article has provided a simple and efficient approach to computing positional uncertainty of landscape features interpreted from aerial imagery and propagating the computed error through the change analysis process.

Accurately assessing large- and small-scale shoreline change has several important implications, such as development, planning, and economics, where the set-back distance of a structure from the coast is directly affected by the observed historical shoreline change. Inaccuracy in the shoreline change assessment may artificially deflate or inflate the legal set-back distance. While exaggerating or overestimating the amount of shoreline change is not as likely to pose a direct threat to the new development, underestimating the change may threaten the development when water-levels rise. The implications of this research are applicable across a wide variety of fields, including urban planning, climate change, glaciology, coastal geomorphology, forestry, and soil geomorphology. We hope this article may inspire further investigation into pragmatic alternatives for uncertainty propagation in shoreline change.

Future research is needed to extend this approach into meso- and macro-tidal environments where the shoreline location can be significantly impacted by fluctuations in water elevation. Accounting for short-term natural variability in shoreline position caused by fluctuating water levels are site-specific and depend on the availability of ancillary water level information, although this correction process is beyond the scope of this article. In addition, future research is needed to extend this approach to linear regression approaches for assessing shoreline change rates. This paper computes the average annual shoreline change rate using the endpoint approach, where the magnitude of change from t_1 to t_2 is divided by the elapsed time between t_1 and t_2 . Extending the ODB approach to linear regression change rates would enable researchers to account for general trends while reducing the impact of single outliers in the shoreline position.

Disclosure statement

No potential conflict of interest was reported by the authors.

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